

Nise 2.13

Enter the following transfer into MATLAB

$$G(s) = \frac{5(s+15)(s+26)(s+72)}{s(s+55)(s^2+5s+30)(s+56)(s^2+27s+52)}$$

Ratio of Factors (zpk) Method

First, use the roots() function to find the poles associate with the two quadratic terms in the denominator.

```
a = roots([1 5 30])
```

```
a = 2x1 complex  
-2.5000 + 4.8734i  
-2.5000 - 4.8734i
```

```
b = roots([1 27 52])
```

```
b = 2x1  
-24.9127  
-2.0873
```

Note that the results are 2x1 **column vectors**.

Collect the real valued poles as a vector.

```
real_poles = [0, -55, -56]
```

```
real_poles = 1x3  
0 -55 -56
```

Note that this generates a 1x3 **row vector**.

How do we concatenate the two three vectors? We could combine them as a set of column vectors,

```
poles_column = [a;  
                b;  
                real_poles']
```

```
poles_column = 7x1 complex  
-2.5000 + 4.8734i  
-2.5000 - 4.8734i  
-24.9127 + 0.0000i  
-2.0873 + 0.0000i  
0.0000 + 0.0000i  
-55.0000 + 0.0000i  
-56.0000 + 0.0000i
```

Or we could concatenate the three vectors as row vectors,

```
poles_row = [a', b', real_poles]
```

```
poles_row = 1x7 complex
```

-2.5000 - 4.8734i -2.5000 + 4.8734i -24.9127 + 0.0000i -2.0873 + 0.0000i ...

Either way can be used with zpk

```
zpk_column= zpk([-15 -26 -72],poles_column,5)
```

```
zpk_column =
```

$$\frac{5 (s+15) (s+26) (s+72)}{s (s+2.087) (s+24.91) (s+55) (s+56) (s^2 + 5s + 30)}$$

Continuous-time zero/pole/gain model.

```
zpk_row= zpk([-15 -26 -72],poles_row,5)
```

```
zpk_row =
```

$$\frac{5 (s+15) (s+26) (s+72)}{s (s+2.087) (s+24.91) (s+55) (s+56) (s^2 + 5s + 30)}$$

Continuous-time zero/pole/gain model.